Graphic Era
HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Even) Semester Regular/Back Examination May 2026

Roll no....226.15.14.........

Name of the Program and semester: B.Tech.(CSE-VIII Sem.)
Name of the Course: Mobile Applications Development
Course Code: TCS-822

Time: 3 hours Maximum Marks: 100

Note:

- *(i)* All the questions are compulsory.
- *(ii)* Answer any two sub questions from a, b and c in each main question.
- *(iii)* Total marks for each question is 20 (twenty).
- *(iv)* Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Describe the role of each layer in the Android architecture, including Linux Kernel, Libraries, Android Runtime (ART), Application Framework, and Applications. (CO-2)

b. What is the significance of MainActivity.java in an Android project? How does it serve as the entry point of an application? (CO-2)

c. What is an Android Virtual Device (AVD)? Explain its significance and the process of creating an AVD in Android Studio. (CO-2)

Q2. (2X10=20 Marks)

a. Describe the process of creating a second activity in an Android app. How can you use Intents to link multiple activities? (CO-2)

b. What is Gesture Detection in Android? How can you implement Swipe, Double Tap, and Pinch Zoom gestures using Gesture Detector? (CO-3)

c. Compare Linear Layout, Relative Layout, Absolute Layout, Table Layout, and Frame Layout with examples. When should each layout be used? (CO-3)

Q3. (2X10=20 Marks)

a. Explain the difference between frame-by-frame animation and tweened animation. How can each be implemented in Android? (CO-2)

b. Compare and contrast sending messages via SMS and email from Android applications. In what scenarios is one preferred over the other? (CO-3)

c. Describe the Android animation framework. Compare property animations, view animations, and drawable animations with suitable examples. (CO-3)

Q4. (2X10=20 Marks)

a. Implement an Android application that downloads data from the internet using AsyncTask. Provide a detailed explanation and code for fetching JSON data via HTTP and updating the UI based on the response. (CO-4)

b. Describe the lifecycle methods of an AsyncTask, including onPreExecute(), doInBackground(), onProgressUpdate(), and onPostExecute(). How are these methods executed and in which thread? (CO-3)

c. Develop an application that displays a user's current location on Google Maps and updates the location as the user moves. Discuss the role of LocationCallback, LocationRequest, and GoogleApiClient. (CO-5)

Q5. (2X10=20 Marks)

a. What are the best practices for testing and optimizing an Android app before release? Include discussions on performance profiling, UI testing, and crash handling. (CO-3)

b. Discuss the importance of application signing in Android. What are the types of keystores used, and how does signing protect the integrity of the application? (CO-2)

c. Discuss how to monitor and respond to user feedback and crash reports using tools like Google Play Console, Firebase Crashlytics, and Google Analytics for Firebase after publishing an app. (CO-2)